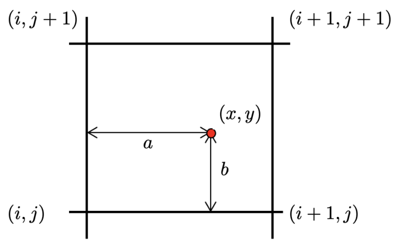
1 The digital image

A digital image is a discrete representation of a continuous 2D function $I(x, y)$. In sensors the analog digital has to be converted by an ADC (analog to digital converter). During exposure the photons are accumulated in the sensor wells, with the accumulated charge being transferred line-by-line to the sense amplifiers which then pass the signal to the ADC. In CMOS sensors each pixel has its own amplifier and ADC.
If we're sampling a signal with a frequency f_s we can only uniquely capture signals with frequencies up to $f_s/2$ (Nyquist frequency). To reconstruct a continuous signal from the sample we could use biliner iterpolation:

$$\hat{f}(x, y) = (1-a)(1-b)F_{i,j} + a(1-b)F_{i+1,j} + abF_{i+1,j+1} + (1-a)bF_{i,j+1}$$



Noise (σ)

- **Additive Gaussian Noise:** This assumes that the noise follows a normal distribution, i.e. $\tilde{F} - F = \sigma \sim \mathcal{N}(0, \tilde{\sigma}^2)$
- **Multiplicative Noise:** Here, the noise follows the intensity of the original signal: $\tilde{F} - F = \sigma = F * c, \quad c \sim \mathcal{N}(b, \tilde{\sigma}^2)$
- **Poisson noise (shot noise):** To model scene-dependent noise, the poisson distribution can be used: $p(k) = \frac{\lambda^k e^{-\lambda}}{k!}$, where λ is the signal average. Note that for large averages, the poisson distribution is similar to a gaussian distribution.
- **Salt-and-Pepper-Noise:** Random black and white pixels are typically caused by faulty sensors or memory errors, hard to describe mathematically, easily removed with median filter.

There are a number of statistics that aim to quantify noise over an image. Among these, the most popular statistics are:

- **Signal-to-Noise-Ratio (SNR):** $\sigma_{SNR} = \frac{E[F]}{E[\sigma]}$. Because of its wide dynamic range, it is often reported in decibel (dB).
- **Peak Signal-to-Noise-Ratio (PSNR):** $\sigma_{SNR} = \frac{F_{max}}{E[\sigma]}$. For most images, F_{max} is 255.

2 Image Segmentation

Classicfication problem where we partition image into disjoint segments.

2.1 Thresholding

Choose a threshold T and classify pixels as foreground or background depending on whether their intensity is above or below T. For color images, use distance in color space instead of intensity to gage similarity.

$$B(x, y) = \begin{cases} 1 & \text{if } I(x, y) > T \\ 0 & \text{else} \end{cases} \qquad \text{(Grayscale)}$$

$$B(x, y) = \begin{cases} 1 & \text{if } \|(I(x, y), (r, g, b)_{\text{ref}})\|_2 > T \\ 0 & \text{else} \end{cases} \qquad \text{(Color)}$$

Instead of using a fixed threshold, we can model the background pixel values as a Gaussian distribution from a set of reference points $c^i_{\text{ref}} \in \mathbb{N}^3$ and then using the Mahalanobis distance to classify pixels:

$$\mu = \frac{1}{N} \sum_{i=1}^{N} c^i_{\text{ref}} \quad \Sigma = \frac{1}{N-1} \sum_{i=1}^{N} (c^i_{\text{ref}} - \mu)(c^i_{\text{ref}} - \mu)^T$$

$$B(x, y) = \begin{cases} 1 & \text{if } (I(x, y) - \mu)^T \Sigma^{-1} (I(x, y) - \mu) > T, \\ 0 & \text{else} \end{cases}$$

As with any classification problem, we can evaluate it using ROC curves. In the script the positive class is the foreground.

$$\text{TPR(Recall)} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad \text{FPR} = \frac{\text{FP}}{\text{FP} + \text{TN}} \quad \text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

2.2 Connected Component Labeling

Connected components are regions in an image where each pixel is connected with paths where each consecutive pixel pair is a neighbour and satisfies some inclusion criteria. The 4-neighbourhood of a pixel are the pixels directly above, below, left and right of it. The 8-neighbourhood also includes the diagonal pixels.

2.3 Region Growing

Iterative process that considers the neighbours of a set of seed pixels and adds them to the region if they satisfy some criteria.

- Select seed point or initial region
- Add neighboring pixels that meet the criteria for inclusion
- Repeat the process until no more pixels can be added

For instance, after each iteration, you can calculate the region's mean and standard deviation, adding a new pixel only if it's mean normalized intensity falls within a specified range of the standard deviation.

2.4 Clustering

We can group pixels in different ways, like intensity similarity, color similarity, texture similarity, intensity+position similarity. Detects sperical clusters only.

2.5 Markov Random Fields

Also want to enforce region constrains like spacial consistency and smooth borders. To achieve this we encode the cost of flipping labels as well as dependencies between pairs of pixels. We're trying to find the Assignment $\mathbf{y} \in \mathbb{N}^{W \times H}$ based on a learnable paramenter θ and some data (e.g. image) with the highest probability:

$$\arg\max P(\mathbf{y}; \theta, \text{data}) = \frac{1}{Z} \prod_{i=1}^{N} p_1(y_i; \theta, \text{data}) \prod_{i,j \in \text{edges}} p_2(y_i, y_j; \theta, \text{data})$$

$$= \arg\min -\log P(\mathbf{y}; \theta, \text{data})$$

$$= \arg\min \sum_i \psi_1(y_i; \theta, \text{data}) + \sum_{i,j \in \text{edges}} \psi_2(y_i, y_j; \theta, \text{data})$$

ψ_1 cost of assigning a label to each pixel, ψ_2 cost of assigning a pair of labels to connected pixels.

Example

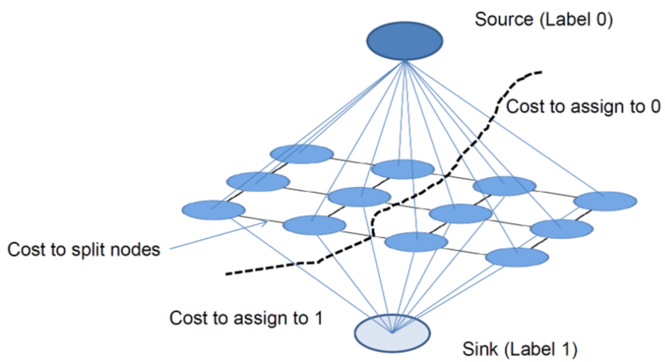
If we want to assign all pixels the label l, the potentials could be:

$$\psi^l_1(y_i; \theta, data) = -\log p(y_i = l | \theta, data)$$

$$\psi^l_2(y_i, y_j; \theta, data) = \begin{cases} 0 & \text{if } y_i = y_j \\ K & \text{else} \end{cases}$$

K is a regularization term that penalizes different labels between connected pixels.

This can be modeled as a graph min-cut problem as follows:



3 Fourier Transform

$$F(u) = \int_{-\infty}^{\infty} f(x)e^{-i2\pi ux}dx,$$

$$f(x) = \int_{-\infty}^{\infty} F(u)e^{i2\pi ux}du$$

$$e^{i2\pi ux} = \cos(2\pi ux) + i\sin(2\pi ux)$$

F(u) holds the amplitude and phase of the corresponding sinusoid. Amplitude/Magnitude: $A(u) = |F(u)| = \sqrt{(\Re\{F(u)\})^2 + (\Im\{F(u)\})^2}$.

Phase:$\phi(u) = \arg F(u) = \tan^{-1}\left(\frac{\Im\{F(u)\}}{\Re\{F(u)\}}\right)$

$$F(u,v) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)e^{-i2\pi(ux+vy)}dx\,dy,$$

$$f(x,y) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} F(u,v)e^{i2\pi(ux+vy)}du\,dv$$

2d basis function:

$$e^{i2\pi(ux+vy)} = \cos(2\pi(ux+vy)) + i\sin(2\pi(ux+vy))$$

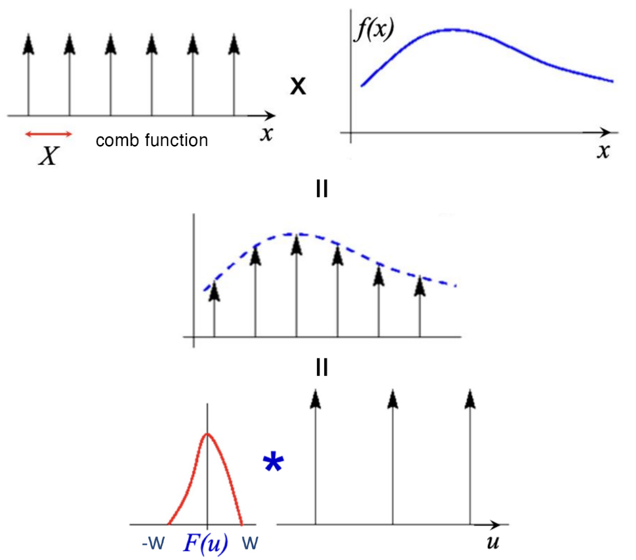
Basis function has maximal real value when $ux + vy \in \mathbb{N}$. Places where this holds form a line in 2d space characterized by normal vector (u, v). The distance between two lines (wavelength) is $\frac{1}{\sqrt{u^2+v^2}}$, frequency is inverse. u and v determine frequency and orientation of the basis function. Pluggin u, v into the Fourier transform of the functiion gives a complex number from which the phase and amplitude can be extracted.

- **Magnitude spectrum** tells you how much of each frequency is present (energy distribution).
- **Phase spectrum** tells you how those frequencies are aligned in space, and this is what gives an image its actual structure.

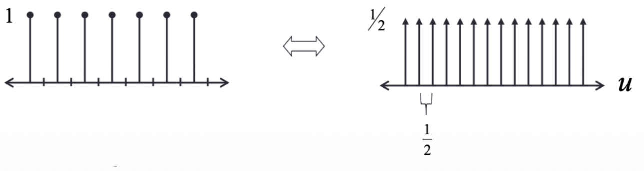
Convolution theorem:

$$f(x,y) * h(x,y) \iff F(u,v)\cdot H(u,v)(\text{(element wise)})$$

4 Sampling



As the comb samples get further apart, the frequency samples get closer together.



Nyquist theorem

$f_{\max}\cdot 2 \le 1/x =$ sampling rate Bc of the convolution theorem it holds: $f_{\text{comb}} \cdot f_{\text{spatial}} = f_{\text{comb}} * F(u) (= f_{\text{frequency}})$

5 Optical Flow

We assume brightness constancy. Let I(x,y,t) be the image brightness at location (x,y) at time t. Then:

$$I(x,y,t) = I(x + \frac{dx}{dt}\delta t, y + \frac{dy}{dt}\delta t, t + \delta t)$$

$$\overset{\text{taylor}}{\approx} I(x,y,t) + \frac{\partial I}{\partial x}\cdot\frac{dx}{dt}\delta t + \frac{\partial I}{\partial y}\cdot\frac{dy}{dt}\delta t + \frac{\partial I}{\partial t}\cdot\delta t$$

$$\implies \frac{\partial I}{\partial x}\frac{dx}{dt} + \frac{\partial I}{\partial y}\frac{dy}{dt} + \frac{\partial I}{\partial t} = 0$$

To linearize I using Taylor expansion we assume the motion is small from frame to frame. Denoting $\frac{\partial I}{\partial x}, \frac{\partial I}{\partial y}, \frac{\partial I}{\partial t}$ as I_x, I_y, I_t respectively, and $\frac{dx}{dt}, \frac{dy}{dt}$ as u, v, the equation is commonly written as $I_x \cdot u + I_y \cdot v + I_t = 0$

5.1 Horn-Schunk Algorithm

Since neighboring pixels tend to move similarly we can add a smoothness constraint.

Global Smoothness

Let u_x denote the change of u in x direction, and so on for u_y, v_x, v_y. We then define the smoothness erro as:

$$e_s := \iint u_x^2 + u_y^2 + v_x^2 + v_y^2 dxdy$$

Denoting $e_c := \iint (I_x u + I_y v + I_t)^2 dxdy$, the algorithm focuses on minimizing the combined error $e := e_c + \lambda e_s$ where λ is a weighting factor. It is minmized using iterative methods with finite differences.

5.2 Lukas-Kanade Algorithm

This algorithm aims to enforce consistency in small patches. We assume a single velocity for all pixels in a patch Ω. And minimize: $\sum_{(x,y)\in\Omega}(I_x(x,y)u + I_y(x,y)v + I_t(x,y))^2$ whose least squares solution is given by solving the system:

$$\begin{bmatrix}\sum I_x^2 & \sum I_x I_y \\ \sum I_x I_y & \sum I_y^2\end{bmatrix}\begin{bmatrix}u \\ v\end{bmatrix} = -\begin{bmatrix}\sum I_x I_t \\ \sum I_y I_t\end{bmatrix}$$

Sometimes there isn't a unique solution to the system, like when a patch contains only an edge moving to the right with no corner visible (gradient in one direction is zero).
The algorithm fails however if the intensity structure in the patch is insufficient (e.g. flat region, edge) or the movement is too large. To mitigate these issues the algorithm is run on increasingly finer scales of the image with the previous coarser scale's flow used as initialization for the next finer scale.

6 Convolution and Filtering

7 General Graphics

Graphics Pipeline Stages:

Modeling Transform	Transforms objects from their local object space to world space by applying translation, rotation, and scaling operations.
Viewing Transform	Transforms objects from world space to camera/view space, positioning them relative to the camera's viewpoint.
Primitive Processing	Processes geometric primitives (triangles, lines, points) and prepares them for the rendering pipeline.
3D Clipping	Clips geometry against the view frustum to remove parts of primitives that fall outside the visible region.
Projection to Screen Space	Projects 3D coordinates onto a 2D screen space using perspective or orthographic projection.
Scan Conversion	Rasterizes geometric primitives into fragments/pixels, determining which pixels are covered by each primitive.
Lighting, Shading, Texturing	Calculates the color of each fragment based on lighting models, applies shading computations, and maps textures onto surfaces.
Occlusion Handling	Determines visibility using depth testing (z-buffering) to ensure that closer objects obscure farther ones.
Display	Outputs the final rendered image to the screen/framebuffer for presentation.

8 Programmer's View

Vertex Processing	Performs per-vertex operations including model transformations, lighting, and flow control.
Rasterization	Converts geometric primitives into fragments; interpolates vertex attributes (e.g., colors, texture coordinates) to generate per-pixel inputs for the fragment shader.
Fragment Processing	Performs per-fragment operations such as shading, texturing, and blending to determine the final pixel color.
Display	Outputs the processed fragments to the framebuffer for final display.

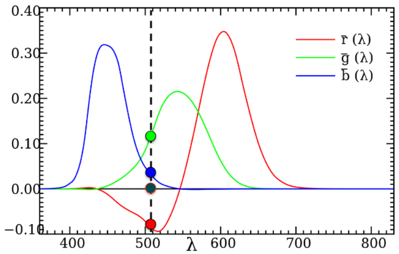
Vertex Shader example

```
uniform mat4 modelViewMatrix;
uniform vec3 lightPosition;
attribute vec3 pos;
attribute vec3 vertexNormal;
varying vec3 lightDirection, normal;
void main() {
  vec4 vertexPosition = modelViewMatrix * pos;
  lightDirection = vec3(lightPosition - vertexPosition);
  normal = vertexNormal;
  gl_Position = vertexPosition;
}
```

9 Light and Color

Light is a electromagnetic radiation at Frequency (wavelength $400nm$ to $700nm$). Spectral Power Distribution:
$P(\lambda) =$ intensity at wavelength λ.
Our eye has 3 types of cones (S (blue),M (green),L (red)) with different sensitivity curves. Eye projects $P(\lambda)$ into 3D subspace using these three basis functions/vectors (two diffrent SPD might look exactly the same to us).

9.1 CIE



$435.8, 546.1$ and $700.0nm$ (arbitrary choosen) where controlled (dimmed/increased) by observers to match a simulus color. Result is seen above, x is frequency and y is the dimmed strength that the participents did. Negativ light is not physically possible, was added bc participents could not recreate this color with just red, green and blue.

$$\text{reference} = \text{blue} + \text{green} - \text{red}$$
$$\text{reference} + \text{red} = \text{blue} + \text{green}$$

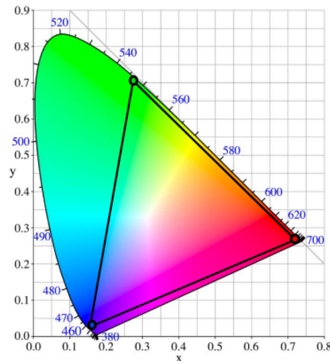
9.2 CIE-XYZ/xyY color space

choosing the $\hat{r}, \hat{g}, \hat{b}$ basis is fairly arbitrary, to avoid negative values we can define a new basis:

$$\begin{pmatrix} \bar{x}(\lambda) \\ \bar{y}(\lambda) \\ \bar{z}(\lambda) \end{pmatrix} = \begin{pmatrix} 2.36 & -0.515 & 0.005 \\ -0.89 & 1.426 & 0.014 \\ -0.46 & 0.088 & 1.009 \end{pmatrix} \begin{pmatrix} \bar{r}(\lambda) \\ \bar{g}(\lambda) \\ \bar{b}(\lambda) \end{pmatrix}$$

$$X = \int_0^\infty P(\lambda)\bar{x}(\lambda)d\lambda$$
$$Y = \int_0^\infty P(\lambda)\bar{y}(\lambda)d\lambda$$
$$Z = \int_0^\infty P(\lambda)\bar{z}(\lambda)d\lambda$$
$$x = \frac{X}{X+Y+Z}$$
$$y = \frac{Y}{X+Y+Z}$$

While Y describes brightness, x and y describe color and give us the CIE chromaticity chart:



- Primary colors are along the edge of the horseshoe
- linear combinations of colors are along straight lines
- linear combinations of 3 colors form a triangle

9.3 CMY color space

- **Passive Systems:** Used for printers (subtractive mixing).
- **Inverse to RGB:**

$$\begin{bmatrix} C \\ M \\ Y \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

- **CMYK:** Adds **Black (K)** to improve contrast and reduce ink usage.

9.4 YIQ Color Space

- **Components:** Luminance (Y), In-phase (I, orange-blue, Humans are very sensitive to this so it got more bandwidth), Quadrature (Q, purple-green, kind of blind in those colors).
- **Usage:** NTSC (US TV standard); advantageous for natural/skin colors.
- **RGB to YIQ Transformation:**

$$\begin{bmatrix} Y \\ I \\ Q \end{bmatrix} = \begin{bmatrix} 0.299 & 0.587 & 0.114 \\ 0.596 & -0.275 & -0.321 \\ 0.212 & -0.523 & 0.311 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

9.5 HSV & HSL/HSB Color Spaces

- **Concept:** User-oriented and intuitive (painters/designers); separates color info (H, S) from intensity (V).
- **Geometry:** Derived by projecting the RGB Cube onto a hexagon (forming a Hexcone).
- **Dimensions:**
  - **Hue (H):** The "base color" (angle on the hexagon, $0° - 360°$).
  - **Saturation (S):** Purity/Richness (radius from center; $0 =$ gray).
  - **Value (V):** Brightness/Lightness (vertical axis).
- **RGB $\rightarrow$ HSV Conversion:** code TODO.

9.6 CIELAB and CIELUV Color Spaces

moving on the horseshoe changes the perceived color difference non-linearly (sometimes tini movement and big change in precived color, sometimes the oposit). To fix this we can define a new color space where euclidean distances correspond to perceived color differences.

9.7 high dinamic range (HDR)

take multiple images at different exposures and combine them to get a higher dynamic range.

10 Miscallaneous

- $e^{i\pi} + 1 = 0$
- $e^{i2\pi ux} = \cos(2\pi ux) + i\sin(2\pi ux)$
- $e^{i2\pi ux} = \cos(2\pi ux) + i\sin(2\pi ux)$
- $e^{-i2\pi ux} = \cos(-2\pi ux) + i\sin(-2\pi ux)$
- $\cos(2\pi ux) = \frac{e^{i2\pi ux} + e^{-i2\pi ux}}{2}$
- $\sin(2\pi ux) = \frac{e^{i2\pi ux} - e^{-i2\pi ux}}{2i}$